

In accordance to Lang (2005), exchange rate volatility is the main driving force of Croatian foreign exchange intervention while the exchange rate level turns out to be insignificant. According to Chmelarova and Schnabl (2006), Croatian foreign exchange intervention manages both day-to-day exchange rate volatilities as well as exchange rate levels. Furthermore, the authors concluded that the pattern of foreign exchange intervention for Croatia confirms a fear of depreciation (with respect to balance sheet effects of the banking sector) more than a fear of appreciation (with respect to export competitiveness). Following the previous research results and empirical data on exchange rate movements, the EUR/HRK exchange rate might be strictly controlled by the Croatian National Bank. Since the Croatian National Bank cannot control the USD/HRK exchange rate at the same time, it's interesting to compare the EUR/HRK and the USD/HRK volatilities pattern.

4. Research data

Financial time series often exhibit volatility clustering, meaning that high volatility periods tend to be followed by high volatility periods and low volatility periods tend to be followed by low volatility periods. In that case, a strong autocorrelation in squared returns or autoregressive conditional heteroscedasticity is present. As a consequence, the least squares estimators are still unbiased but inefficient. The estimates of the variances are biased, thus invalidating the tests of significance, and the obtained results are dubious (see for example Erjavec and Cota, 2007). In order to resolve the problem and obtain estimator efficiency, as a method of estimation several ARCH type models has been employed. As in most empirical finance literature, the variable to be modeled is the daily exchange rate return which is the first difference of the natural logarithm of the exchange rate and is given by the following equation:

$$r_t = \ln \left(\frac{S_t}{S_{t-1}} \right) \quad (1)$$

Where r_t is the daily exchange rate return and S_t and S_{t-1} denote the Croatian National Bank (CNB) mid-point exchange rate of the EUR versus the HRK and the USD versus the HRK at the current day and pre-

vious day, respectively. The data span from 1st January 1997 to 30th September 2015 is used as a data sample for modeling the daily exchange rate return. Table 1 shows the descriptive statistics for the observed variables.

Table 1 Descriptive statistics for the daily exchange rates return of the EUR and the USD versus the HRK

Variable	r_{t-EUR}	r_{t-USD}
Mean	2.34E-05	4.36E-05
Median	1.44E-05	-7.19E-05
Maximum	0.010004	0.035861
Minimum	-0.008836	-0.035029
Std. Dev	0.001345	0.006620
Skewness	0.081265	0.018182
Kurtosis	6.346057	4.453776
Jarque-Bera	2199.145	414.4102
Probability	0.000000	0.000000
Number of observations	4703	4703

Source: Authors' calculation

A key feature of exchange rate returns is that the distribution of returns is fat tailed. That is, the probability density function of exchange rate returns appears to be leptokurtic, so it is more peaked at the center and has fatter tails compared to that of the normal distribution. Numerically, the kurtosis coefficient is found to be greater than 3, which characterizes kurtosis of a normally distributed random variable. In addition, exchange rate returns tend to be slightly skewed either to the left or to the right which is not consistent with returns being normally distributed. The positive value of skewness indicates that data are skewed to the right referring to a depreciation of the currency (HRK). As can be seen in Table 1, the exchange rate returns series (r_t) exhibits significant values of skewness and kurtosis, and therefore normality assumptions are not met. In accordance to abovementioned alternative, distributions have to be used as a basis for modeling exchange rate returns, such as the Student-t or Generalized Error distribution (GED) rather than the normal distribution which takes into account the phenomenon of leptokurtosis and skewness in the probability density function.